# Introduction to Complete Segal Spaces

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In this talk we introduced complete Segal spaces, which are an important model for higher categories. The primary source for this topic is still the work of Rezk [Re01], which is also the original source.

Before we move on to talk about complete Segal spaces, we give a quick review some basic definitions and notation regarding spaces and simplicial spaces, which already introduced in our first talk ([Sh17]).

## Spaces

We denote with $\mathcal{S}$ the category of spaces (simplicial sets). An object $X \in \mathcal{S}$ is a functor $X\colon \Delta^{op} \to \mathsf{Set}$ and morphisms are natural transformations between such functors. We think of $X$ as a collection of sets, indexed by the natural numbers, with maps between them. Intuitively, we have a set $X_0$ of points, a set $X_1$ of edges between these points, a set $X_2$ of faces (2-dimensional objects), and so on. The maps between these sets indicate how they are glued together. This idea is made precise by defining the geometric realization adjunction between $\mathcal{S}$ and $\mathsf{Top}$ (as is done in section 3 of [Sh17]).

**Remark 1.1.** For every $n \in \mathbb{N}$ we get a representable functor, which we denote by $\Delta^n$. By the Yoneda Lemma we get a bijection of sets

$$\mathrm{Hom}_{\mathcal{S}}(\Delta^n, X) \cong X_n$$

for every $X \in \mathcal{S}$.

One of the remarkable facts about $\mathcal{S}$ is that it is enriched over itself. For every $X, Y \in \mathcal{S}$ we define the space $\mathrm{Map}(X, Y) \in \mathcal{S}$ as

$$\mathrm{Map}_{\mathcal{S}}(X, Y)_n := \mathrm{Hom}_{\mathcal{S}}(\Delta^n \times X, Y)$$

which makes the category $\mathcal{S}$ is the prototypical example of a simplicial category.

**Remark 1.2.** Note that at level 0 we just regain $Hom_{\mathcal{S}}(X, Y)$.

One of the crucial features of $\mathcal{S}$ is that it can be assigned a simplicial model structure, namely the Kan model structure. Recall that in the Kan model structure weak equivalences are weak equivalences of the corresponding topological

spaces (under geometric realization), fibrations are maps satisfying the Kan extension property and cofibrations are just inclusions. For more details on this see section 4 of [Sh17].

## Simplicial Spaces

A simplicial space $X$ is a simplicial object in the category of spaces, that is a functor $X\colon \Delta^{op} \to \mathcal{S}$. We denote as $s\mathcal{S}$ the category of simplicial spaces with natural transformations between them, thus for each $m \in \mathbb{N}$ we have that $X_m \in \mathcal{S}$, i.e. the collection of $m$-simplices is a space.

$s\mathcal{S}$ can be enriched over $\mathcal{S}$ (see Section 5 of [Sh17]), making $s\mathcal{S}$ a simplicial category. Note that there is a full embedding $i : \mathcal{S} \hookrightarrow s\mathcal{S}$, defined by $i(X)_n = X$. We also call these *constant simplicial spaces* and we can think of it as a "vertical embedding" of spaces into simplicial spaces). We call the image of $\Delta^n$ again just $\Delta^n$. This is the standard way we embed spaces into simplicial spaces.

There is also a *horizontal embedding*, $j : \mathcal{S} \hookrightarrow s\mathcal{S}$, which is defined as $j(X)_{nl} = X_n$, thus at each level our simplicial space is just discrete (a set). This embedding is non-standard and we will use it only in this following specific case. We can again embed $\Delta^k$ but under the vertical embedding. We denote the image of that as $F(k) \in s\mathcal{S}$. Concretely, $F(k)$ is a discrete simplicial space such that

$$F(k)_n = Hom_{\Delta}([n],[k])$$

**Remark 2.1.** A Yoneda-type argument shows that $F(k)$ represents the $k$-th space functor, i.e.

$$\mathrm{Map}_{s\mathcal{S}}(F(k), X) \simeq X_k$$

is an isomorphism in $\mathcal{S}$.

The category of simplicial spaces comes with a very convenient model structure, *the Reedy model structure*. It defines weak equivalences (cofibrations) as level-wise weak equivalences (cofibrations). Importantly, this makes $s\mathcal{S}$ into a cofibrantly-generated, simplicial model category (see Section 5 of [Sh17] for details).

Our goal is to put certain conditions on simplicial spaces such that under those conditions they have a suitable categorical behavior. Intuitively, we want to think of $X_0$ as "the space of objects" and $X_1$ as the "space of morphisms" and it should satisfy the following two conditions:

1. The higher spaces $X_2, X_3, ...$ should somehow "encode" how to compose "morphisms"
2. If we fix an $n$ then the higher simplices in $X_n$ (i.e. $X_{n1}, X_{n2}, ...$) should encode "homotopies" of $X_{n0}$.

In order to get a better feeling how to precisely formulate those conditions, in the next section we will first discuss the *classification diagram* of a category.

## Classifying spaces and diagrams

Before we define classifying spaces, we will first review the notion of a *nerve*

**Definition 3.1.** Given a category $C$, we can define its *nerve* $NC$ as the space given by

$$NC_m = \mathrm{Hom}_{\mathsf{Cat}}([m], C)$$

Thus $m$-simplices are sequences of $m$ composable morphisms in $C$.

**Remark 3.2.** The nerve construction provides a full embedding $N\colon \mathsf{Cat} \hookrightarrow \mathcal{S}$. Note that equivalences of categories are sent to weak equivalences in $\mathcal{S}$, but non equivalent categories can be sent to weakly equivalent spaces (the non-equivalent map of categories $[0] \to [1]$ is sent to $\Delta^0 \to \Delta^1$ which is certainly a Kan equivalence). Thus we might be interested in having a notion that actually reflects weak equivalences rather than just preserve them.

The geometric realization $|NC| \in \mathsf{Top}$ is called the *classifying space* of the category $C$.

In [Re01], Charles Rezk introduced a more general construction, called the *classifying diagram* (also known as the *Rezk nerve*), which provides a full embedding $\mathcal{N}\colon \mathsf{Cat} \to s\mathcal{S}$. This construction actually fixed the deficit as it turns out that two categories $C$ and $D$ are equivalent if and only if their respective classifying diagrams $\mathcal{N}C$ and $\mathcal{N}D$ are weakly equivalent. Before we can define it though we will need to remind ourselves of one last definition.

**Definition 3.3.** For each category $C$ we define the core of $C$, denoted $C^{core}$, as the maximal sub-groupoid that lies inside $C$. Concretely, $C^{core}$ has the same objects as $C$ but the maps have to be isomorphims.

**Definition 3.4.** For each category $C$ we define the classifying diagram as follows:

$$\mathcal{N}C_m := N((C^{[m]})^{core})$$

for each $m \in \mathbb{N}$, where $C^{[m]} = Fun([m], C)$ i.e. the category of functors from $[m]$ to $C$.

Let us break down what this definition is implying. More explicitely, the category $(C^{[m]})^{core}$ is described as follows: objects are sequences of $m$ composable morphisms of $C$, and maps between two objects are diagrams in $C$ like the one below, where all the squares commute and the vertical arrows are isomorphisms.

$$\begin{tikzcd}
\bullet_0 \arrow[r] \arrow[d, "\simeq"'] & \bullet_1 \arrow[r] \arrow[d, "\simeq"'] & \bullet_2 \arrow[r] \arrow[d, "\simeq"'] & \cdots \arrow[r] & \bullet_m \arrow[d, "\simeq"'] \\
\bullet_0 \arrow[r] & \bullet_1 \arrow[r] & \bullet_2 \arrow[r] & \cdots \arrow[r] & \bullet_m.
\end{tikzcd} \tag{3.1}$$

The nerve of this category is the simplicial set whose $n$-simplices are $n$ (vertically) composable diagrams of the form 3.1. To summarize, an element in $(\mathcal{N}C_m)_n$ is an $(m, n)$- grid of composable morphisms of $C$, where all the vertical arrows are isomorphisms.

**Example 3.5.** Let us look at some lower level scenarios:

0 $\mathcal{N}C_0$ is the space whose 0-simplices are objects of $C$, 1-simplices are isomorphisms between objects of $C$, 2-simplices are pairs of composable isomorphisms in $C$ and so on.

1 $\mathcal{N}C_1$ is the space whose 0-simplices are the maps of $C$, 1-simplices commutative squares where vertical maps are isomorphims and so on.

We are finally in a position to give a precise definition for our earlier intuition using the example case above as motivation. Given the way we defined classifying spaces, it is not difficult to verify that it has the following property:

$$\mathcal{N}C_m \xrightarrow{\quad\varphi_m\quad} \mathcal{N}C_1 \times_{\mathcal{N}C_0} \cdots_{\mathcal{N}C_0} \times \mathcal{N}C_1 \tag{3.2}$$

is an isomorphism of spaces for all $m \in \mathbb{N}_{\geq 2}$ (the map $\varphi_n$ will be discussed in the next section). This is simply asserting that in $\mathcal{N}C$ we have a genuinely unique way to compose maps, or better, given a sequence of $m$ composable edges there is a unique way to fill in all the cells up to dimension $m$. In the terminology of [Re16], we have unique extension along a spine. In a Segal spaces we want composition to be unique "up to homotopy", and in order to achieve this we will have to relax condition 3.2 and ask instead for the maps $\varphi_m$ to be weak equivalences.

## Segal spaces

For $m \geq 2$, let

$$G(m) = F(1) \coprod_{F(0)} F(1) \coprod_{F(0)} ... \coprod_{F(0)} F(1)$$

where there are $m$ factors of $F(1)$. There is a natural map $G(m) \to F(m)$, which is the identity at the level of objects. Because of the way $G(m)$ embeds into the simplex $F(m)$ we often call $G(m)$ **"the spine"** of $F(m)$.

For every simplicial space $X$, the map $G(m) \to F(m)$ induces a map of spaces:

$$\varphi_m : X_m \to X_1 \times_{X_0} ... \times_{X_0} X_1$$

where there are $m$ factors of $X_1$.

**Definition 4.1.** An object $W \in s\mathcal{S}$ is called a *Segal space* if it is Reedy fibrant and for every $m \in \mathbb{N}_{\geq 2}$ the spine map introduced above

$$W_m \xrightarrow{\quad\varphi_m\quad} W_1 \times_{W_0} \cdots_{W_0} \times W_1 \tag{4.1}$$

is a weak equivalence.

**Remark 4.2.** Since this map was induced by the inclusion $G(m) \hookrightarrow F(m)$, which is a cofibration and $W$ is already Reedy fibrant, we conclude that $\varphi_m$ is always a Kan fibration. We could have thus defined Segal space as a Reedy fibrant object $W \in s\mathcal{S}$ such that for all $m \in \mathbb{N}_{\geq 2}$, the map $\varphi_m$ is a trivial fibration.

**Example 4.3.** The Rezk nerve of a category is our main example of a Segal space. We have already observed that the condition on the $\varphi_m$'s is true in a stronger form for $\mathcal{N}C$ . Proving that $\mathcal{N}C$ is Reedy fibrant requires us to check some lifting condition which we will omit here ([Re01, Lemma 3.9]).

# Category Theory for Segal Spaces

In this section we start doing category theory in the context of Segal spaces. In particular, we discuss mapping spaces and the notion of an equivalence.

**Definition 5.1.** We say that $x$ is an object in $W$, and we write $x \in W$, if $x$ is a 0-simplex in $W_0$. In other words $ObjW = W_{00}$.

**Remark 5.2.** We will soon construct a category out of $W$ where these are actually the objects of an honest category.

**Definition 5.3.** For $x_0, x_1 \in W$ we define the *mapping space* $\mathcal{M}\mathrm{ap}_W(x_0, x_1)$ as the space identified by the following pullback diagram

$$\begin{tikzcd} \mathcal{M}\mathrm{ap}_W(x_0, x_1) \arrow[r] \arrow[d] & W_1 \arrow[d] \\ \{(x_0, x_1)\} \arrow[r] & W_0 \times W_0. \end{tikzcd} \tag{5.1}$$

**Remark 5.4.** Note that the map $W_1 \to W_0 \times W_0$ is a Kan fibration. Thus the space $\mathcal{M}\mathrm{ap}_W(x_0, x_1)$ is actually a Kan complex.

**Example 5.5.** For each object $x \in W$, we have the map $s_0 x \in \mathcal{M}\mathrm{ap}(x, x)$, which we normally denote as $id_x$. Later (Proposition 5.8) we will justify this notation.

There is no point in having a mapping space, if we don't have any composition. However, there is no way to just compose maps in a Segal space. Fortunately, there is still something we can do but for that we need extended mapping spaces.

**Definition 5.6.** For $x_0, \ldots, x_m \in W$ we define $\mathcal{M}\mathrm{ap}_W(x_0, \ldots, x_m)$ as the pullback

$$\begin{tikzcd} \mathcal{M}\mathrm{ap}_W(x_0, \ldots x_m) \arrow[r] \arrow[d] & W_m \arrow[d] \\ \{(x_0, \ldots, x_m)\} \arrow[r] & W_0 \times \cdots \times W_0. \end{tikzcd} \tag{5.2}$$

Note that $\mathcal{M}\mathrm{ap}_W(x_0, x_1)$ is a subspace of $W_1$ (intuitively, it keeps track of all the edges between $x_0$ and $x_1$, of the homotopies between them, and of the

compositions of these homotopies). Similarly, $\mathcal{M}\mathrm{ap}_W(x_0, \dots x_m)$ is a subspace of $W_m$. Exploiting the map $\varphi_m$ we obtain a trivial fibration

$$\mathcal{M}\mathrm{ap}_W(x_0, \dots, x_m) \xrightarrow{\simeq} \mathcal{M}\mathrm{ap}_W(x_{m-1}, x_m) \times \cdots \times \mathcal{M}\mathrm{ap}_W(x_0, x_1). \quad (5.3)$$

Now to the punchline of this section. The maps 5.3 allow us to define composition in $W$ in such a way that the composite of two (composable) maps is well defined up to homotopy. Indeed, let $f \in \mathcal{M}\mathrm{ap}_W(x_0, x_1)$ and $g \in \mathcal{M}\mathrm{ap}_W(x_1, x_2)$. Then in the following diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & \mathcal{M}\mathrm{ap}_W(x_0, x_1, x_2) \arrow[d, "\simeq"] \\ \{(g, f)\} \arrow[r] & \mathcal{M}\mathrm{ap}_W(x_1, x_2) \times \mathcal{M}\mathrm{ap}_W(x_0, x_1). \end{tikzcd} \quad (5.4)$$

it is easily seen that $A$ is contractible, hence there is a "unique choice for a composition, up to homotopy". Now there is a boundary map $\mathcal{M}\mathrm{ap}_W(x_0, x_1, x_2) \to \mathcal{M}\mathrm{ap}_W(x_0, x_2)$, which gives us our "indirect composition map":

$$\mathcal{M}\mathrm{ap}_W(x_0, x_1) \times \mathcal{M}\mathrm{ap}_W(x_1, x_2) \xleftarrow{\simeq} \mathcal{M}\mathrm{ap}_W(x_0, x_1, x_2) \to \mathcal{M}\mathrm{ap}_W(x_0, x_2)$$

We will often abuse notation and denote any choice of composition in $\mathcal{M}\mathrm{ap}_W(x_0, x_2)$ as $g \circ f$.

Since $\mathcal{M}\mathrm{ap}_W(x_0, x_1)$ is a space, the following definition is natural.

**Definition 5.7.** We say that two elements $f, g \in \mathcal{M}\mathrm{ap}_W(x_0, x_1)$ are *homotopic* and we write $f \sim g$ if they belong to the same path component of $\mathcal{M}\mathrm{ap}_W(x_0, x_1)$.

It turns out that composition is associative and unitary up to homotopy. More precisely, we have

**Proposition 5.8.** [Re01, Proposition 5.4] For three maps $f \in \mathcal{M}\mathrm{ap}_W(x_0, x_1)$, $g \in \mathcal{M}\mathrm{ap}_W(x_1, x_2)$ and $h \in \mathcal{M}\mathrm{ap}_W(x_2, x_3)$ we have that $(h \circ g) \circ f \sim h \circ (g \circ f)$ and that $f \circ \mathrm{id}_{x_0} \sim f \sim \mathrm{id}_{x_1} \circ f$.

We can thus define a *homotopy category*.

**Definition 5.9.** For each Segal space $W$ we define the homotopy category $\mathrm{Ho}W$ whose objects are the objects of $W$ and for $x, y \in W$,

$$\hom_{\mathrm{Ho}W}(x, y) = \pi_0 \mathcal{M}\mathrm{ap}_W(x, y)$$

The following definition is now natural (and crucial).

**Definition 5.10.** A point $f \in \mathcal{M}\mathrm{ap}_W(x_0, x_1)$ is a *homotopy equivalence* if there exists $g, h \in \mathcal{M}\mathrm{ap}_W(x_0, x_1)$ such that $f \circ g \sim \mathrm{id}_{x_1}$ and $h \circ f \sim \mathrm{id}_{x_0}$, that is if $f$ is invertible in $\mathrm{Ho}W$.

Being a homotopy equivalence is "homotopically invariant" in a strong sense:

**Proposition 5.11.** [Re01, Lemma 5.8] If $f \in W_1$ is a homotopy equivalence, then every $g \in X_1$ belonging to the same path component of $f$ is also a homotopy equivalence.

To say it differently, in a path component of $W_1$ either all points are homotopy equivalences or none is. Thus we can define the subspace of $W_1$ consisting only of homotopy equivalences. We name this space $W_{hoequiv}$. As we just showed that if one point lies in $W_{hoequiv}$ then it whole path-component has to be in $W_{hoequiv}$ or in other words that the inclusion map $W_{hoequiv} \to W_1$ is (-1)-connected.

Let us take a step back and consider what we have achieved so far. Thanks to the Segal condition 4.1 we were able to build a composition for points of $W_1$, which is well defined, unitary and associative up to homotopy. We therefore have a solid notion of equivalence for elements of $W_1$. We called such invertible elements homotopy equivalences.

At this point we have two notions of "sameness" available for two elements $x, y \in W$. We can either say that they are equivalent if they lie in the same path component of $W_0$ or alternatively if there exists a homotopy equivalence between them, that is if there is a homotopy equivalence $f \in \mathcal{M}\mathrm{ap}_W(x, y)$. Clearly the first implies the second, but in general opposite is not true. Making them coincide will be the task of the completeness condition.

## The completeness condition

In the last section we defined the space $W_{hoequiv}$. Note that the map $s_0 \colon W_0 \to W_1$ necessarily factors through $W_{\mathrm{hoequiv}}$, since $s_0 x = \mathrm{id}_x$ is a homotopy equivalence for any $x \in W_0$.

**Definition 6.1.** A *complete Segal space* is defined to be a Segal space $W$ for which the map $s_0 \colon W_0 \to W_{\mathrm{hoequiv}}$ is a weak equivalence.

As a consequence of Proposition 5.11 a Segal space $W$ is a complete Segal space if and only if the connected components of $\mathrm{id}_x$ for $x \in W$ exhaust all the homotopy equivalences.

We now describe another way to interpret this property. Unwinding the definition of a weak equivalence, we see that a map $f \in \mathcal{M}\mathrm{ap}_W(x, y)$ is a homotopy equivalence if and only if for the diagram below all the dotted arrows and all connecting 2-faces and the filling 3-simplex exist. Here the two dotted arrows in the front will play the role of a post-inverse ($hf \simeq id_x$) and the one dotted arrow in the back plays the role of a pre-inverse ($fg \simeq id_y$) and the three-cell

in the middle tells us that all the data is coherent.

$$\begin{tikzcd}
 & \bullet & \\
\bullet \arrow[ur, dashed, "h"] \arrow[rr, dashed, "g"] \arrow[dr, "\mathrm{id}_y"'] & & \bullet \arrow[ul, "\mathrm{id}_x"'] \arrow[dl, "f"] \\
 & \bullet \arrow[uu, dashed] &
\end{tikzcd} \tag{6.1}$$

The undotted arrows exactly to correspond to the image of the map

$$i : W_1 \to W_1 \times_{W_0} W_1 \times_{W_0} W_1$$

which maps $f$ to $i(f) = (id_x, f, id_y)$. What we basically just stated is the following: $f \in W_1$ lies in $W_{hoequiv}$ if and only if it's image into $W_1 \times_{W_0} W_1 \times_{W_0} W_1$ can be lifted to $W_3$.

The explanation we just gave is just the universal condition for a pullback square, thus we have:

$$\begin{tikzcd}
W_0 \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
 & W_{\text{hoequiv}} \arrow[r] \arrow[d] & W_3 \arrow[d] \\
 & W_1 \arrow[r] & W_1 \times_{W_0} W_1 \times_{W_0} W_1.
\end{tikzcd} \tag{6.2}$$

The completeness condition is asking for the dotted map in 6.2 to be a weak equivalence. The general philosophy here is that for two points in $W$, a condition (being connected by a homotopic equivalence) that is tested at level 3, becomes equivalent to some information (path connectedness in $W_0$) already available at level 0.

**Example 6.2.** As expected, the Rezk nerve $\mathcal{N}C$ of a category $C$ is a complete Segal space. Intuitively, this is true because between two objects $c, c' \in C$ there exists an isomorphism if and only if there are arrows filling the diagram below, with the vertical arrows being isomorphisms.

$$\begin{tikzcd}
c \arrow[r] \arrow[d, "\simeq"'] & c' \arrow[d, "\simeq"'] \\
c \arrow[r] & c'.
\end{tikzcd} \tag{6.3}$$

A more rigorous proof can be found in [Re01, Proposition 6.1].

## Closed model category structures

Making use of the Bousfield localisation procedure (see [Sh17, Theorem 6.1] or [Re01, Proposition 9.1]) we can construct a simplicial closed model structure on

$s\mathcal{S}$ such that the fibrant objects are exactly the Segal spaces (resp. complete Segal spaces).

**Theorem 7.1.** There is a model structure on $s\mathcal{S}$ called the *Segal space model structure*, which satisfies the following conditions:

- An object $W$ is fibrant if it is a Segal space
- A map $g : C \to D$ is a weak equivalence if for Segal space $W$, the map
$$Map(D, W) \to Map(C, W)$$
is a Kan equivalence.
- Cofibrations are just inclusions
- Fibrations are maps which satisfy the lifting conditions with repect to acyclic cofibrations.
- Weak equivalences (fibrations) between Segal spaces are just Reedy weak equivalences (fibrations).

We build this model structure by localizing the Reedy model structure with respect to cofibrations of the form:

$$G(m) \hookrightarrow F(m) \tag{7.1}$$

where $m \geq 2$.

**Theorem 7.2.** There is a model structure on $s\mathcal{S}$ called the *complete Segal space model structure*, which satisfies the following conditions:

- An object $W$ is fibrant if it is a complete Segal space
- A map $g : C \to D$ is a weak equivalence if for every complete Segal space $W$, the map
$$Map(D, W) \to Map(C, W)$$
is a Kan equivalence.
- Cofibrations are just inclusions
- Fibrations are maps which satisfy the lifting conditions with repect to acyclic cofibrations.
- Weak equivalences (fibrations) between complete Segal space are just Reedy weak equivalences (fibrations).

We build this model structure by localizing the Segal space model structure with respect to the cofibration of the form:

$$F(0) \hookrightarrow F(3) \coprod_{F(1) \coprod_{F(0)} F(1) \coprod_{F(0)} F(1)} F(1) \tag{7.2}$$

## References


[Re01] C. Rezk, *A model for the homotopy theory of homotopy theory*, Trans. Amer. Math.Soc., 353(2001), no. 3, 973-1007.

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